

Introduction to ML

A subset of artificial intelligence known as machine learning focuses primarily on the creation of algorithms that enable a computer to independently learn from data and previous experiences.

Arthur Samuel first used the term "machine learning" in 1959.

MACHINE LEARNING EXPLAINED

Machine learning is in the form of data driven systems. The focus of AI is to develop intelligent systems by using data-driven approaches, where data is used as an input to develop intelligent models. The models can then be used to predict new inputs. Thus, the aim of machine learning is to learn a model or set of rules from the given dataset automatically so that it can predict the unknown data correctly.

As humans take decisions based on an experience, computers make models based on extracted information patterns in the input data and then use these data-filled models for prediction and to make decisions.

Often, the quality of data determines the quality of experience and, therefore, the quality of the learning system. In statistical learning, the relationship between the input x and output y is modeled as a function in the form $y = f(x)$. Here, f is the learning function that maps the input x to output y .

Learning of function f is the crucial aspect of forming a model in statistical learning. Learning program summarizes the raw data in a model. Formally stated, a model is an explicit description of patterns within the data in the form of:

1. Mathematical equation
2. Relational diagrams like trees/graphs
3. Logical if/else rules, or
4. Groupings called clusters

In summary, a model can be a formula, procedure or representation that generates data decisions. For example, a model can be helpful to examine whether a given email is spam or not. The point is that the model is generated automatically from the given data.

Models of computer systems are equivalent to human experience. Experience is based on data. Humans gain experience by various means. They gain knowledge by rote learning. They observe others and imitate them. Humans gain a lot of knowledge from teachers and books. We learn many things by trial and error. Once the knowledge is gained, when a new problem is encountered, humans search for similar past situations and then formulate the heuristics and use that for prediction.

But, in systems, experience is gathered by these steps:

1. Collection of data
2. Once data is gathered, abstract concepts are formed out of that data. Abstraction is used to generate concepts. This is equivalent to humans' idea of objects, for example, we have some idea about how an elephant looks like.
3. Generalization converts the abstraction into an actionable form of intelligence. It can be viewed as ordering all possible concepts. So, generalization involves ranking of concepts, inferencing from them and formation of heuristics, an actionable aspect of intelligence. Heuristics are educated guesses for all tasks.
4. Heuristics normally works! But, occasionally, it may fail too. It is not the fault of heuristics as it is just a 'rule of thumb'. The course correction is done by taking evaluation measures. Evaluation checks the thoroughness of the models and to-do course correction, if necessary, to generate better formulations.

Machine Learning in relation to other fields

Machine learning and Artificial Intelligence:

The AI happened due to development of data driven systems. The aim is to find relations and regularities present in the data. Machine learning is the sub branch of AI, whose aim is to extract the patterns for prediction. It is a broad field that includes learning from examples and other areas like reinforcement learning. The relationship of AI and machine learning is shown in Figure 1.3. The model can take an unknown instance and generate results.

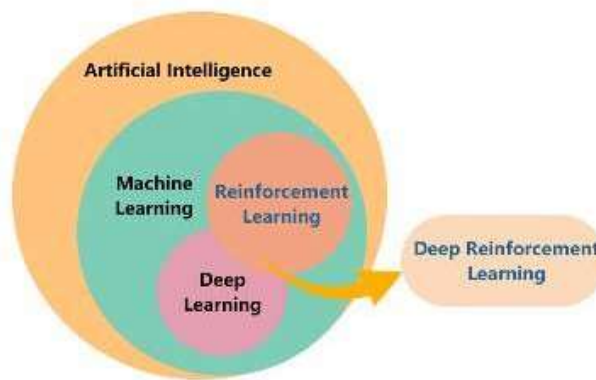


Figure 1.3: Relationship of AI with Machine Learning.

Deep learning is a sub branch of machine learning. In deep learning, the models are constructed using neural network technology. Neural networks are based on the human neuron models. Many neurons form a network connected with the activation functions that trigger further neurons to perform tasks.

Machine Learning, Data Science, Data Mining, and Data Analytics

Data science is an 'Umbrella' term that encompasses many fields. Machine learning starts with data. Therefore, data science and machine learning are interlinked. Machine learning is a branch of data science. Data science deals with gathering data for analysis. It is a broad field that includes.

Big Data: Data Science about collection of data. Big data is a field of data science that deals with data's following characteristics:

1. Volume: Huge amount of data is generated by big companies like Facebook, Twitter
2. Variety: Data is available in variety of forms like images, videos, and in different formats,
3. Velocity: It refers to the speed at which the data is generated and processed.

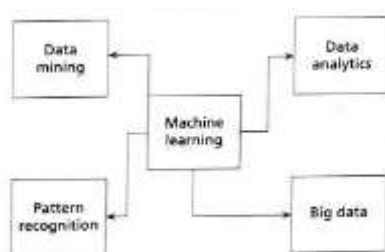
Big data is used by many machine learning algorithms for applications such as language translation and image recognition. Big data influences the growth of subjects like Deep learning. Deep learning is a branch of machine learning that deals with constructing models using neural network

Data Mining Data mining's original genesis is in the business. Like while mining the earth one gets into precious resources, it is often believed that unearthing of the data produces hidden information that otherwise would have eluded the attention of the management. Nowadays, many consider that data mining and machine learning are the same. There is no difference between these fields except that data mining aims to extract the hidden patterns that are present in the data, whereas, machine learning aims to use it for prediction.

Data Analytics: Another branch of data science is data analytics. It aims to extract useful knowledge from crude data. There are different types of analytics. Predictive data analytics is used for making predictions. Machine learning is closely related to this branch of analytics and shares almost all algorithms.

Pattern Recognition: It is an engineering field. It uses machine learning algorithms to extract the features for pattern analysis and pattern classification. One can view pattern recognition as a specific application of machine learning.

These relations are summarized in the figure below.



Machine Learning and Statistics

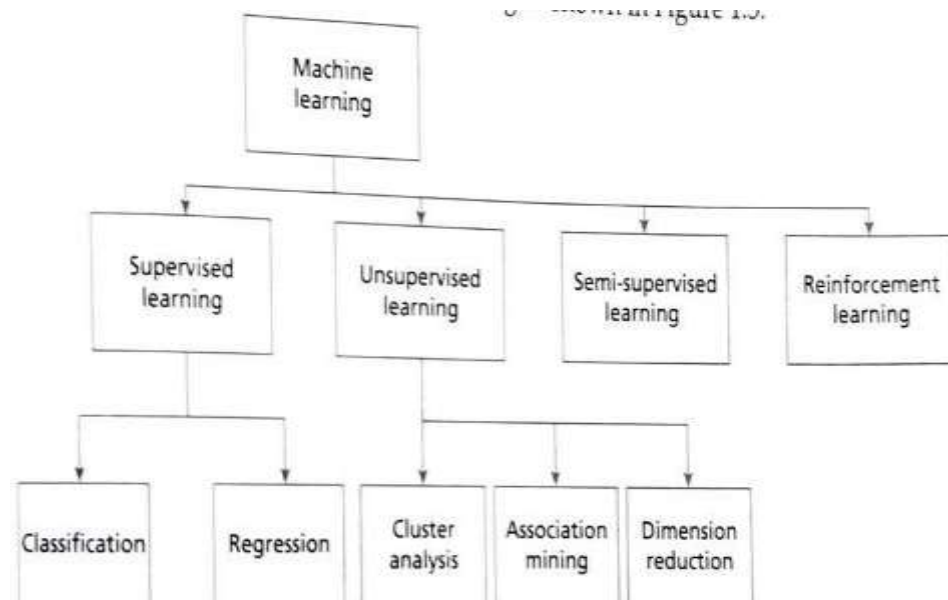
Statistics is a branch of mathematics that has a solid theoretical foundation regarding statistical learning. Like machine learning (ML), it can learn from data. But the difference between statistics and ML is that statistical methods look for regularity in data called patterns. Initially, statistics sets a hypothesis and performs experiments to verify and validate the hypothesis in order to find relationships among data.

Statistics requires knowledge of the statistical procedures and the guidance of a good statistician. It is mathematics intensive and models are often complicated equations and involve many assumptions.

Machine learning ,comparatively , has less assumptions and requires less statistical knowledge. But it often requires interaction with various tools to automate the process of learning.

Types of Machine Learning.

There are 4 types of machine learning.

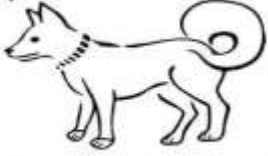


Labeled and Unlabelled Data Data is a raw fact. Normally, data is represented in the form of a table. Data also can be referred to as a data point, sample, or an example. Each row of the table represents a data point. Features are attributes or characteristics of an object. Normally, the columns of the table are attributes. Out of all attributes, one attribute is important and is called a label. Label is the feature that we aim to predict. Thus, there are two types of data - labeled and unlabelled.

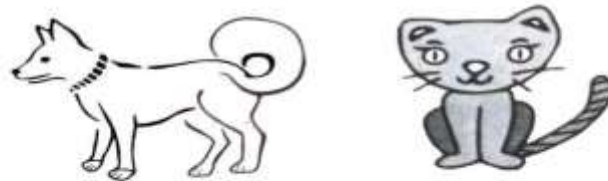
Labeled Data To illustrate labeled data, let us take one example dataset called Iris flower dataset or Fisher's Iris dataset. The dataset has 50 samples of Iris - with four attributes, length and width of sepals and petals. The target variable is called class. There are three classes - Iris setosa, Iris virginica, and Iris versicolor.

Table 1.1: Iris Flower Dataset

S.No.	Length of Petal	Width of Petal	Length of Sepal	Width of Sepal	Class
1.	5.5	4.2	1.4	0.2	Setosa
2.	7	3.2	4.7	1.4	Versicolor
3	7.3	2.9	6.3	1.8	Virginica

Input	Label
	dog
	Cat

(a)



(b)

Figure 1.6: (a) Labelled Dataset (b) Unlabelled Dataset

A dataset need not always be numbers. It can be images or video frames. Deep neural networks can handle images with labels. In the following Figure 1.6, the deep neural network takes images of dogs and cats with labels for classification.

Supervised Learning

Supervised algorithms use labelled dataset. As the name suggests, there is a supervisor or teacher component in supervised learning. A supervisor provides labelled data so that the model is constructed and generates test data.

In supervised learning algorithms, learning takes place in two stages. In layman terms, during the first stage, the teacher communicates the information to the student that the student is supposed to master. The student receives the information and understands it. During this stage, the teacher has no knowledge of whether the information is grasped by the student.

This leads to the second stage of learning. The teacher then asks the student a set of questions to find out how much information has been grasped by the student. Based on these questions the student is tested, and the teacher informs the student about his assessment. This kind of learning is typically called supervised learning.

Supervised learning has two methods:

1. Classification
2. Regression

Classification

Classification is a supervised learning method. The input attributes of the classification algorithms are called independent variables. The target attribute is called a label or dependent variable. The relationship between the input and target variable is represented in the form of a structure which is called a classification model. So, the focus of classification is to predict the 'label' that is a discrete form (a value from the set of finite values). An example is shown in Figure 1.7 where a classification algorithm takes a set of labelled data images such as dogs and cats to construct a model that can later be used to classify an unknown test image data.

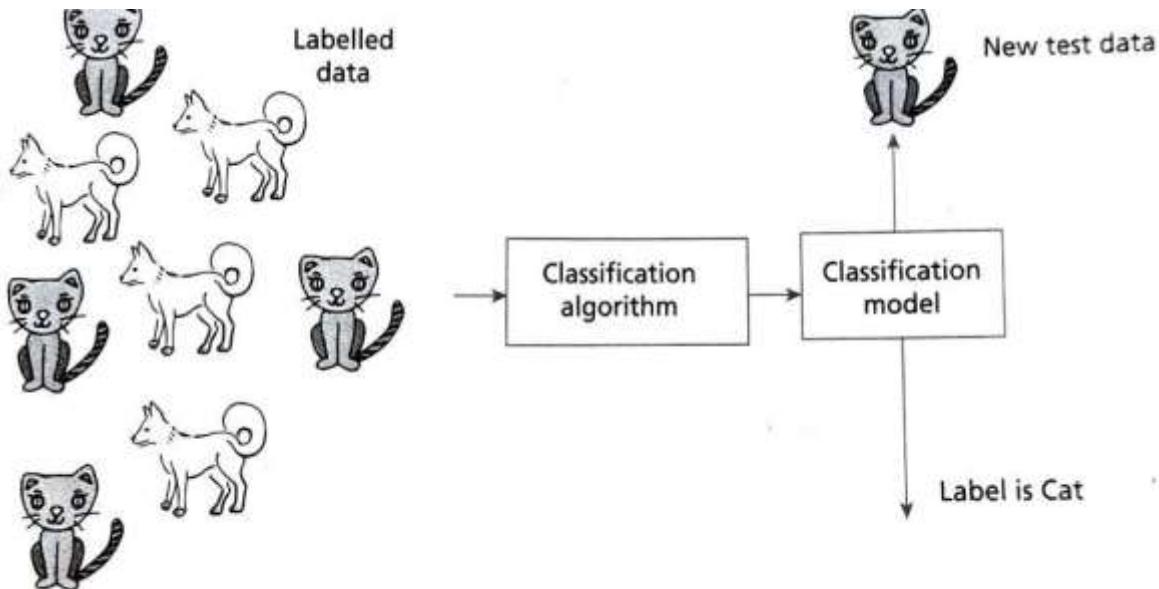


Figure 1.7: An Example Classification System

In classification, learning takes place in two stages. During the first stage, called the training stage, the learning algorithm takes a labelled dataset and starts learning. After the training set, samples are processed and the model is generated. In the second stage, the constructed model is tested with a test or unknown sample and assigned a label. This is the classification process.

This is illustrated in the above Figure 1.7. Initially, the classification learning algorithm learns with the collection of labelled data and constructs the model. Then, a test case is selected, and the model assigns a label.

Some of the key algorithms of classification are:

- Decision Tree
- Random Forest
- Support Vector Machines
- Naïve Bayes
- Artificial Neural Network and Deep Learning networks like CNN

Regression Models

Regression models, unlike classification algorithms, predict continuous variables like price. In other words, it is a number. A fitted regression model is shown in Figure 1.8 for a dataset represent weeks input x and product sales y .

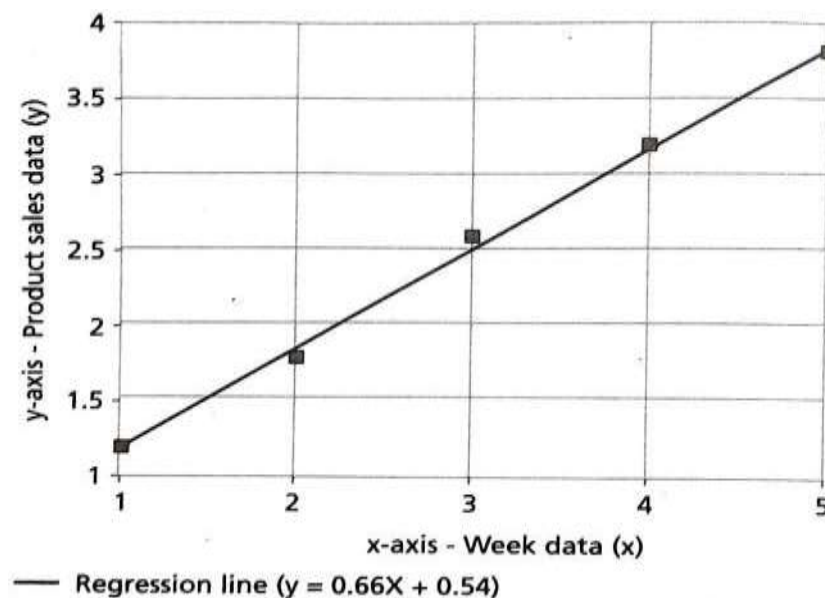


Figure 1.8: A Regression Model of the Form $y = ax + b$

The regression model takes input x and generates a model in the form of a fitted line of the form $y = f(x)$. Here, x is the independent variable that may be one or more attributes and y is the dependent variable. In Figure 1.8, linear regression takes the training set and tries to fit it with a line- product sales $= 0.66 * \text{Week} + 0.54$. Here, 0.66 and 0.54 are all regression coefficients that are learnt from data. The advantage of this model is that prediction for product sales (y) can be made for unknown week data (x). For example, the prediction for unknown eighth week can be made by substituting x as 8 in that regression formula to get y .

Both regression and classification models are supervised algorithms. Both have a classification the concepts of training and testing are applicable to both. What is the difference between variables and regression models (The main difference is that regression models predict continuous such as product price, while classification concentrates on assigning labels such as class).

Here, the program is supplied with objects, but no labels are defined. The algorithm itself observes the examples and recognizes patterns based on the principles of grouping. Grouping is done in ways that similar objects form the same group.

Cluster Analysis and Dimensional reduction algorithms are examples of Unsupervised algorithms.

Cluster Analysis

Cluster analysis is an example of unsupervised learning. It aims to group objects into disjoint clusters or groups. Cluster analysis clusters objects based on their attributes. All the data objects of the partitions are similar in some aspects and vary from the data objects in the other partitions significantly.

Some of the examples of clustering processes are - image detection of abnormal growth in a medical image and determining clusters of signatures, segmentation of a region of interest in an image, detection of abnormal growth in a medical image.

An example of clustering scheme is shown in Figure 1.9 where the clustering algorithm takes a set of dogs and cats images and groups it as two clusters-dogs and cats. It can be observed that the samples belonging to a cluster are similar and samples are different radically across clusters.

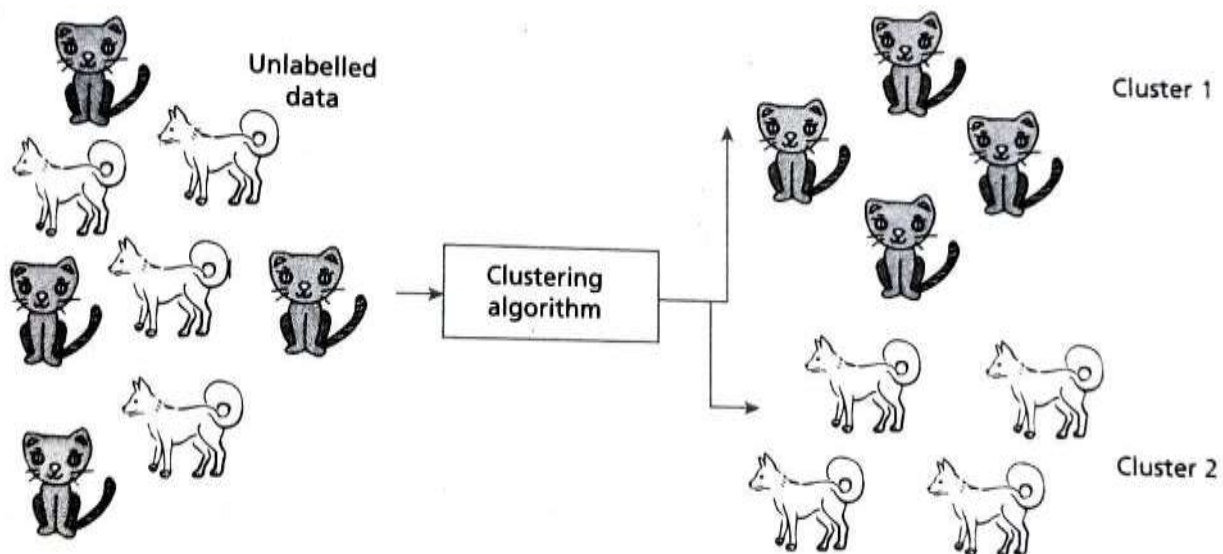


Figure 1.9: An Example Clustering Scheme

Some of the key clustering algorithms are:

- k-means algorithm
- Hierarchical algorithms

Dimensionality Reduction

Dimensionality reduction algorithms are examples of unsupervised algorithms. It takes a higher dimension data as input and outputs the data in lower dimensional by taking advantage of the variance of the data. It is a task of reducing the data set with few features without losing the generality .

Table 1.2

Table 1.2: Differences between Supervised and Unsupervised Learning

S.No.	Supervised Learning	Unsupervised Learning
1	There is a supervisor component	No supervisor component
2	Uses Labelled data	Uses Unlabelled data
3	Assigns categories or labels	Performs grouping process such that similar objects will be in one cluster

Semi-supervised Learning

There are circumstances where the dataset has a huge collection of unlabelled data and some labelled data. Labelling is a costly process and difficult to perform by the humans. Semi-supervised algorithms use unlabelled data by assigning a pseudo-label. Then, the labelled and pseudo-labeled dataset can be combined.

Reinforcement Learning

Reinforcement learning mimics human beings. Like human beings use ears and eyes to perceive the world and take actions, reinforcement learning allows the agent to interact with the environment to get rewards. The agent can be human, animal, robot, or any independent program. The rewards enable the agent to gain experience. The agent aims to maximize the reward.

The reward can be positive or negative (Punishment). When the rewards are more, the behavior gets reinforced and learning becomes possible.

Consider the following example of a Grid game as shown in Figure 1.10.

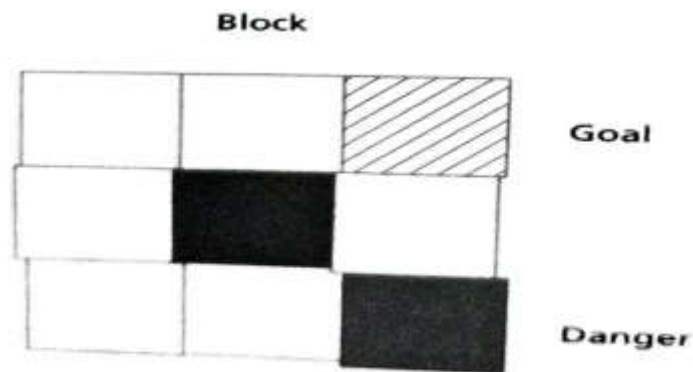


Figure 1.10: A Grid game

In this grid game, the gray tile indicates the danger black is a block and the tile with diagonal lines is the goal. The aim is to start, say from bottom-left grid, using the actions left, right, top and bottom to reach the goal state

To solve this sort of problem, there is no data. The agent interacts with the environment to get experience. In the above case, the agent tries to create a model by simulating many paths and finding rewarding paths. This experience helps in constructing a model.

It can be said in summary, compared to supervised learning, there is no supervisor or Reinforcement algorithms are sequential decisions need to be taken to reach the final decision. Therefore, reward-based, goal-oriented algorithms.

CHALLENGES OF MACHINE LEARNING

What are the challenges of machine learning? Let us discuss about them now.

Problems that can be Dealt with Machine Learning

Computers are better than humans in performing tasks like computation. For example, while calculating the square root of large numbers, an average human may blink but computers can display the result in seconds. Computers can play games like chess, GO, and even beat professional players of that game.

However, humans are better than computers in many aspects like recognition. But, deep learning systems challenge human beings in this aspect as well. Machines can recognize human faces in a second. Still, there are tasks where humans are better as machine learning systems still require quality data for model construction. The quality of a learning system depends on the quality of data. This is a challenge. Some of the challenges are listed below:

1. Problems-Machine learning can deal with the 'well-posed' problems where specifications are complete and available. Computers cannot solve "ill-posed" problems.

Consider one simple example (shown in Table 1.3):

Input (x_1, x_2)	Output (y)
1, 1	1
2, 1	2
3, 1	3
4, 1	4
5, 1	5

Can a model for this test data be multiplication? That is, $y = x_1 * x_2$, Well! It is true! But, this is equally true that y may be $y = x_1 / x_2$, Or $y = x_1(\text{pow}(x_2))$. So, there are three functions that fit the data. This means that the problem is ill-posed. To solve this problem, one needs more examples to Check the model. Puzzles and games that do not have sufficient specification may become an ill-posed problem and scientific computation has many ill-posed problems.

2. Huge data - This is a primary requirement of machine learning data is a challenge. A quality data means it should be large and should not have data problems such as missing data or incorrect data.

3. High computation power -With the availability of Big Data, the computational resource requirement has also increased. Systems with Graphics Processing Unit (GPU) or even Tensor Processing Unit (TPU) are required to execute machine learning algorithms. Also, machine learning tasks have become complex and hence time complexity has increased, and that can be solved only with high computing power.

4. Complexity of the algorithms The selection of algorithms, describing the algorithms, application of algorithms to solve machine learning tasks, and comparison of algorithms have become necessary for machine learning or data sCientists now. Algorithms have become a big topic of discussion and it is a challenge for machine learning professionals to design, select, and evaluate optimal algorithms.

5. Bias/Variance - Variance is the error of the model. This leads to a problem called bias/variance tradeoff. A model that fits the training data correctly but fails for test data. In general, lack of generalization is called overfitting. The reverse problem is called underfitting where the model fails for training data but has good generalization. Overfitting and underfitting are great challenges for machine learning algorithms.

MACHINE LEARNING PROCESS

The emerging process model for the data mining solutions for business organizations is CRISP-DM. Since machine learning is like data mining, except for the aim, this process can be used for machine learning. CRISP-DM stands for Cross Industry Standard Process- Data Mining. This process involves six steps. The steps are listed below in Figure 1.11.

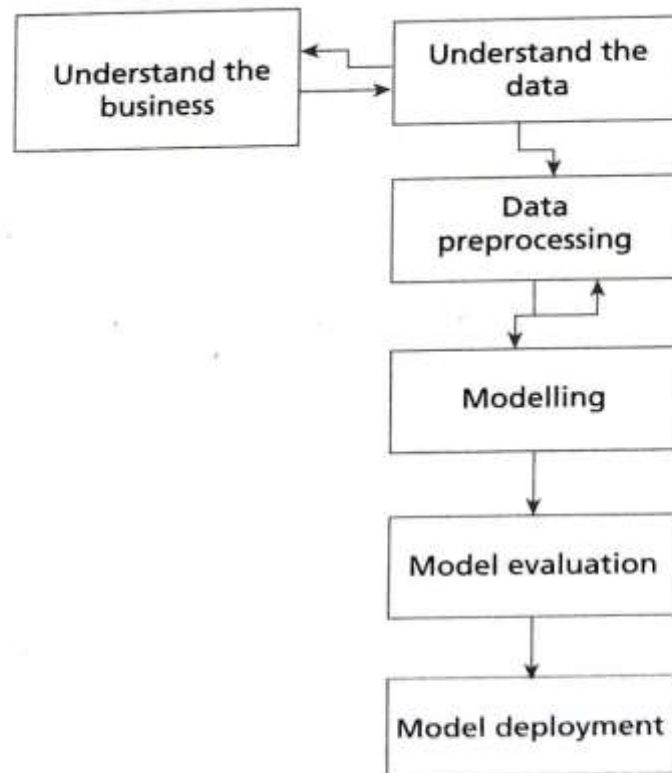


Figure 1.11: A Machine Learning/Data Mining Process

1. Understanding the business - this step involves understanding the objectives and requirements of the business organization. Generally, a single data mining algorithm is enough for giving the solution. This step also involves the formulation of the problem statement for the data mining process.
2. Understanding the data - It involves the steps like data collection, study of the characteristics of the data, formulation of hypothesis, and matching of patterns to the selected hypothesis.
3. Preparation of data - This step involves producing the final dataset by cleaning the raw data and preparation of data for the data mining process. The missing values may cause problems during both the training and testing phases. Missing data forces classifiers to produce inaccurate results. This is a perennial problem for the classification models. Hence suitable strategies should be adopted to handle the missing data.
4. Modelling - This step plays a role in the application of data mining algorithm for the data to obtain a model or pattern.

5. Evaluate-This Step involves the evaluation of the data mining results using statistical analysis and visualization methods, The performance of the classifier is determined by evaluating the accuracy of the classifier. The process of classification is a fuzzy issue. For example, classification of emails requires extensive domain knowledge and requires domain experts. Hence, performance of the classifier is very crucial.
6. Deployment - This step involves the deployment of results of the data mining algorithm to improve the existing process or for a new situation.

MACHINE LEARNING APPLICATIONS

Machine Learning technologies are used widely now in different domains. Machine learning applications are everywhere! One encounters many machine learning applications in day-to-day life. Some applications are listed below:

1. Sentiment analysis - This is an application of natural language processing (NLP) where the words of documents are converted to sentiments like happy, sad, and angry which are captured by emoticons effectively. For movie reviews or product reviews, five stars or one star are automatically attached using sentiment analysis programs.
2. Recommendation systems - These are systems that make personalized purchases possible. For example, Amazon recommends users to find related books or books bought by people who have the same taste like you, and Netflix suggests shows or related movies of your taste. The recommendation systems are based on machine learning.
3. Voice assistants -Products like Amazon Alexa, Microsoft Cortana, Apple Siri, and Google Assistant are all examples of voice assistants. They take speech commands and perform tasks. These chatbots are the result of machine learning technologies.
4. Technologies like Google Maps and those used by Uber are all examples of machine learning which offer to locate and navigate shortest paths to reduce time.

Understanding Data

Data is available in different data sources like flat files, databases, or data warehouses. It can be an operational data or a non-operational data. Operational data is the one that is encountered in normal business procedures and processes.

For example, daily sales data is operational on the other hand, non-operational data is the kind of data that is used for decision making.

Data by itself is meaningless. It has to be processed to generate any information. A string bytes is meaningless. Only when a label is attached, like the height of students of a class, the data becomes meaningful.

Processed data is called information that includes patterns, associations, or relationships among data. For example, sales data can be analyzed to extract information like which product was sold larger in the last quarter of the year

Elements of Big Data

Data whose volume is less and can be stored and processed by a _mall-scale computer is called small data. These data are collected from several sources, and integrated and processed by a small-scale computer. Big data, on the other hand, is a larger data whose volume is much larger than 'small data' and is characterized as follows:

- Volume - Since there is a reduction in the cost of storing devices, there has been a tremendous growth of data. Small traditional data is measured in terms of gigabytes (GB) and terabytes (TB), but Big Data is measured in terms of petabytes (PB) and exabytes (EB). One exabyte is 1 million terabytes.
- Velocity - The fast arrival speed of data and its increase in data volume is noted as velocity. The availability of IoT devices and Internet power ensures that the data is arriving at a faster rate. Velocity helps to understand the relative growth of big data and its accessibility by users, systems and applications.
- Variety- The variety of Big Data includes:
 - ★ Form- There are many forms of data. Data types range from text, graph, audio, video to maps. There can be composite data too, where one media can have many other sources of data, for example, a video can have an audio song.
 - ★ Function- These are data from various sources like human conversations, transaction records, and old archive data.
 - ★ Source of data - This is the third aspect of variety. There are many sources of data. Broadly, the data source can be classified as open/public data, social media data and multimodal data.

Some of the other forms of Vs that are often quoted in the literature as characteristics of Big data are:

- **Veracity of data** - Veracity of data deals with aspects like conformity to the facts, truthfulness, believability, and confidence in data. There may be many sources of error such as technical errors, typographical errors, and human errors. So, veracity is one of the most important aspects of data.
- **Validity** - Validity is the accuracy of the data for taking decisions or for any other goals that are needed by the given problem.
- **Value** - Value is the characteristic of big data that indicates the value of the information that is extracted from the data and its influence on the decisions that are taken based on it.

These 6V's are helpful to characterize the big data. The data quality of the numeric attributes is determined by factors like precision, bias, and accuracy. Precision is defined as the closeness of repeated measurements. Often, standard deviation is used to measure the precision. Bias is a systematic result due to erroneous assumptions of the algorithms or procedures. Accuracy is the degree of measurement of errors that refers to the closeness of measurements to the true value of the quantity. Normally, the significant digits used to store and manipulate indicate the accuracy of the measurement.

Types of Data

In Big Data, there are three kinds of data. They are structured data, unstructured data, and semi-structured data.

Structured Data

In structured data, data is stored in an organized manner such as a database where it is available in the form of a table. The data can also be retrieved in an organized manner using tools like SQL.

The structured data frequently encountered in machine learning are listed below:

Record Data : A dataset is a collection of measurements taken from a process. We have a collection of objects in a dataset and each object has a set of measurements. The measurements can be arranged in the form of a matrix. Rows in the matrix represent an object and can be called as entities, cases, or records. The columns of the dataset are called attributes, features, or fields. The table is filled with observed data. Also, it is better to note the general jargons that are associated with the dataset. Label is the term that is used to describe the individual observations.

Data Matrix : It is a variation of the record type because it consists of numeric attributes. The standard matrix operations can be applied on these data. The data is thought of as points or vectors in the multidimensional space where every attribute is a dimension describing the object.

Graph Data: It involves the relationships among objects. For example, a web page can refer to another web page. This can be modeled as a graph. The nodes are web pages and the hyperlink is an edge that connects the nodes.

Ordered Data : Ordered data objects involve attributes that have an implicit order among them.

The examples of ordered data are:

1. Temporal data - It is the data whose attributes are associated with time. For example, the customer purchasing patterns during festival time is sequential data. Time series data is a special type of sequence data where the data is a series of measurements over time.
2. Sequence data - It is like sequential data but does not have time stamps. This data involves the sequence of words or letters. For example, DNA data is a sequence of four characters -ATGC.
3. Spatial data - It has attributes such as positions or areas. For example, maps are spatial data where the points are related by location.

Unstructured Data

Unstructured data includes video, image, and blog data. It is estimated audio. It also includes textual documents, programs, that 80% of the data are unstructured data.

Semi-Structured Data

Semi-structured data are partially structured and partially unstructured. These include data like VML/JSON data, RSS feeds, and hierarchical data.

Data Storage and Representation

Once the dataset is assembled, it must be stored in a structure that is suitable for data analysis. The goal of data storage management is to make data available for analysis. There are different approaches to organize and manage data in storage files and systems from flat file to data Warehouses.

Some of them are listed below:

Flat Files: These are the simplest and most commonly available data source. It is also the cheapest way of organizing the data. These flat files are the files where data is stored in plain ASCII or EBCDIC format. Minor changes of data in flat files affect the results of the data mining algorithms. Hence, flat file is suitable only for storing small dataset and not desirable if the dataset becomes larger.

Some of the popular spreadsheet formats are listed below:

- CSV files - CSV stands for comma-separated value files where the values are separated by commas. These are used by spreadsheet and database applications. The first row may have attributes and the rest of the rows represent the data.
- TSV files - TSV stands for Tab separated values files where values are separated by Tab. Both CSV and TSV files are generic in nature and can be shared. There are many tools like Google Sheets and Microsoft Excel to process these files.

Database System: It normally consists of database files and a database management system (DBMS). Database files contain original data and metadata. DBMS aims to manage data and improve operator performance by including various tools like database administrator, query processing, and transaction manager. A relational database consists of sets of tables. The tables have rows and columns. The columns represent the attributes and rows represent tuples. A tuple corresponds to either an object or a relationship between objects. A user can access and manipulate the data in the database using SQL.

Different types of databases are listed below:

1. A **transactional database** is a collection of transactional records. Each record is a transaction. A transaction may have a time stamp, identifier and a set of items, which may have links to other tables.
2. **Time-series databases** store time related information like log files where data is associated with a time stamp. This data represents the sequences of data, which represent values or events obtained over a period (for example, hourly, weekly or yearly) or repeated time span. Observing sales of products continuously may yield a time-series data.
3. **Spatial databases** contain spatial information in a raster or vector format . Raster format are either bitmaps or pixel maps. On the other hand , the vector format can be used to store maps as maps use basic geometric primitives like points, lines, etc

World Wide Web (WWW) It provides a diverse, worldwide online information Source. The objective of data mining algorithms is to mine interesting patterns of information present that can be used to represent data that needs to be shared across the platforms.

XML (eXtensible Markup Language): It is both human and machine interpretable data format real-time constraints.

Data Stream: It is dynamic data, which flows in and out of the observing environment. characteristics of data stream are huge volume of data, dynamic, fixed order movement, and real time constraints.

RSS (Really Simple Syndication) It is a format for sharing instant feeds across services.

JSON (JavaScript Object Notation) It is another useful data interchange format that is often used for many machine learning algorithms.

BIG DATA ANALYTICS AND TYPES OF ANALYTICS

Data analysis and data analytics are terms that are used interchangeably to refer to the same Concept. However, there is a subtle difference. Data analytics is a general term and data analysis is a part of it.

Data analytics refers to the process of data collection, preprocessing, and analysis. It deals with the complete cycle of data management.

Data analysis is just analysis and is a part of data analytics. It takes historical data and does the analysis. Data analytics, instead, more on the future and helps in predicting future trends.

There are four types of data analytics:

1. Descriptive analytics
2. Diagnostic analytics
3. Predictive analytics
4. Prescriptive analytics

Descriptive Analytics : It is about describing the main features of the data. After data collection is done, descriptive analytics deals with the collected data and quantifies it. It is often stated that analytics is essentially statistics.

There are two aspects of statistics - Descriptive and Inference.

Descriptive analytics: only focuses on the description part of the data and not the inference part.

Diagnostic Analytics: It deals with the question - Why?. This is also known as causal analysis as it aims to find out the cause and effect of the events.

For example, if a product is not selling diagnostic analytics aims to find out the reason. There may be multiple reasons and associated effects are analyzed as part of it.

Predictive Analytics: It deals with the future. It deals in the future given this data?. This involves the application of algorithms to identify the patterns to predict the future.

Prescriptive Analytics: It is about finding the best course of action for the business organizations. Prescriptive analytics goes beyond prediction and helps in decision making by giving a set of actions. It helps the organizations to plan better for the future and to mitigate the risks that are involved.

BIG DATA ANALYSIS FRAMEWORK

Big data framework is a layered architecture. Such an architecture has many advantages such as genericness. A 4-layer architecture has the following layers:

1. Data connection layer

2. Data management layer
3. Data analytics later
4. Presentation layer

Data Connection Layer: It has data ingestion mechanisms and data connectors. Data ingestion means taking raw data and importing it into appropriate data structures. It performs the tasks of the ETL process. By ETL, it means extract, transform and load operations.

Data Management Layer: It performs preprocessing of data. The purpose of this layer is to allow parallel execution of queries, and read, write and data management tasks. There may be many schemes that can be implemented by this layer such as data-in-place, where the data is not moved at all, or constructing data repositories such as data warehouses and pull data on-demand mechanisms.

Data Analytic Layer It has many functionalities such as statistical tests, machine learning algorithms to understand, and construction of machine learning models. This layer implements many model validation mechanisms too.

Presentation Layer It has mechanisms such as dashboards, and applications that display the results of analytical engines and machine learning algorithms.

Thus, the Big Data processing cycle involves data management that consists of the following steps.

1. Data collection
2. Data preprocessing
3. Applications of machine learning algorithm
4. Interpretation of results and visualization of machine learning algorithm

This is an iterative process and is carried out on a permanent basis to ensure that data is Suitable for data mining.

Data Collection

The first task of gathering datasets is the collection of data. It is often estimated that most of the time is spent for collection of good quality data. A good quality data yields a better result. It is often difficult to characterize a Good data.

'Good data' is one that has the following properties:

1. Timeliness - The data should be relevant and not stale or obsolete data.
2. Relevancy -The data should be relevant and ready for the machine learning or data mining algorithms. All the necessary information should be available and there should be no bias in the data.

3. Knowledge about the data - The data should be understandable and interpretable, and should be self-sufficient for the required application as desired by the domain knowledge engineer.

Broadly, the data source can be classified as open/public data, social media data and multimodal data.

1. Open or public data source - It is a data source that does not have any stringent copyright rules or restrictions. Its data can be primarily used for many purposes. Government census data are good examples of open data:

- Digital libraries that have huge amount of text data as well as document images
- Scientific domains with a huge collection of experimental data like genomic data and biological data
- Healthcare systems that use extensive databases like patient databases, health insurance data, doctors' information, and bioinformatics information

2. Social media -It is the data that is generated by various social media platforms like Twitter, Facebook, YouTube, and Instagram. An enormous amount of data is generated by these platforms.

3. Multimodal data It includes data that involves many modes such as text, video, audio and mixed types. Some of them are listed below:

- Image archives contain larger image databases along with numeric and text data.
- WWW has a huge amount of data that is distributed on the Internet . These are heterogeneous in nature.

Data Preprocessing

In real world, the available data is 'dirty'.

1. Incomplete data
2. Outlier data
3. Inaccurate data
4. Data with missing values
5. Duplicate data
6. Data with inconsistent values.

Data preprocessing improves the quality of the data mining techniques. The raw data must be preprocessed to give accurate results. The process of detection and removal of errors in data, is called **data cleaning**

Data wrangling means making the data processable for machine learning algorithms. Some of the data errors include human errors such as typographical errors or incorrect measurement and structural errors like improper data formats. Data errors can also arise from omission and duplication of attributes. Noise is a random component and involves distortion of a value or introduction of spurious objects.

Patient ID	Name	Age	Date of Birth (DoB)	Fever	Salary
1.	John	21		Low	-1500
2.	Andre	36		High	Yes
3.	David	5	10/10/1980	Low	" "
4.	Raju	136		High	Yes

It can be observed that data like Salary Andre, = is incomplete data. The DoB of patients, John and Raju, is the missing data. The age of David is recorded as 5 but his DoB indicates it is 10/10/1980. This is called inconsistent data.

Inconsistent data occurs due to problems in conversions, inconsistent formats, and difference in units. Salary for John is -1500. It cannot be less than 0. It is an instance of noisy data. Outliers are data that exhibit the characteristics that are different from other data and have very unusual values. The age of Raju cannot be 136. It might be a typographical error.

Missing Data Analysis:

The primary data cleaning process is missing data analysis. Data cleaning routines attempt to fill up the missing values, smoothen the noise while identifying the outliers. This enables data mining to avoid overfitting of the models.

The procedure that are given below can solve the problem of missing data

1. Ignore the tuple - A tuple with missing data, especially the class label is ignored. This method is not effective when the percentage of the missing values increases.
2. Fill in the values manually - Here, the domain expert can analyze the data tables and carry out the analysis and fill in the values manually. But, this is time consuming and may not be feasible for larger sets.
3. A global constant can be used to fill in the missing attributes. The missing values may be 'Unknown' or be 'Infinity'. But, some data mining results may give spurious results by analyzing these labels.
4. The attribute value may be filled by the attribute value. Say, the average income can replace a missing value.
5. Use the attribute mean for all samples belonging to the same class. Here, the average value replaces the missing values of all tuples that fall in this group.
6. Use the most possible value to fill in the missing value. The most probable value can be obtained from other methods like classification and decision tree prediction.

Removal of Noisy or Outlier Data

Noise is a random error or variance in a measured value. It can be removed by using binning, which is a method where the given data values are sorted and distributed into equal frequency bins. The bins are also called buckets. The binning method then uses the neighbor values to smooth the noisy data.

Some of the techniques commonly used are 'smoothing' by means' where the mean of the bin removes the values of the bins, 'smoothing by bin medians' where the bin median replaces the bin values, and 'smoothing by bin boundaries' where the bin value is replaced by the closest bin boundary. The maximum and minimum values are called bin boundaries. Binning methods may be used as a discretization technique.

Data Integration & Data Transformation

Data Integration involves routines that merge data from multiple sources into a single data source . So this may lead to redundant data. The main goal of data integration is to detect and remove redundancies that arise from integration.

Data transformation routines perform operations like normalization to improve the performance of the data mining algorithms. It is necessary to transform data so that it can be processed.

Normalization is one such technique which is scaled to fit in a range(0-1) to improve the performance of the data mining algorithm. Some of the normalization procedures used are

- Min-Max
- z-Score

Min-Max Procedure : It is a normalization technique where each variable V is normalized by its difference with the minimum value divided by the range to a new range 0-1.

The formula is given as

$$v'_i = \frac{v_i - \min}{\max - \min} (\text{new_max} - \text{new_min}) + \text{new_min}$$

Here max-min is the range . Min & max are the minimum & maximum of the given data , new max and new min are the minimum and maximum to target range ,say 0 & 1

z-Score Normalization : This procedure works by taking the difference between the field value and mean value and by scaling this difference by standard deviation of the attribute.

$$Z = \frac{\text{raw score} - \text{mean}}{\text{standard deviation}}$$

$$Z = \frac{X - \mu}{\sigma}$$

Where

X is the raw score

μ is the mean

σ is the standard deviation

Data Reduction

Data reduction reduces data size but produces the same results. There are different ways in which data reduction can be carried out such as data aggregation , feature selection and dimensionality reduction.

Descriptive Statistics

Descriptive statistics is a branch of statistics that does dataset summarization. It is used to summarize and describe data. Descriptive statistics are just descriptive and do not go beyond that

Data visualization is a branch of study that is useful for investigating the given data. Mainly, the plots are useful to explain and present data to customers.

Descriptive analytics and data visualization techniques help to understand the nature of the data, which further helps to determine the kinds of machine learning or data mining tasks that can be applied to the data. This step is often known as Exploratory Data Analysis (EDA). The focus of EDA is to understand the given data and to prepare it for machine learning algorithms. EDA includes descriptive statistics and data visualization

The descriptive statistics with the **fundamental concepts of datatypes**.

- 1. Dataset and Data Types:** A dataset can be assumed to be a collection of data objects. The data objects may be records, points, vectors, patterns, events, cases, samples or observations. These records contain many attributes. An attribute can be defined as the property or characteristics of an object.

Table 2.2: Sample Patient Table

Patient ID	Name	Age	Blood Test	Fever	Disease
1.	John	21	Negative	Low	No
2.	Andre	36	Positive	High	Yes

For example, consider the following database shown in sample Table 2.2. Every attribute should be associated with a value. This process is called measurement. The type of attribute determines the data types, often referred to as measurement scale types.

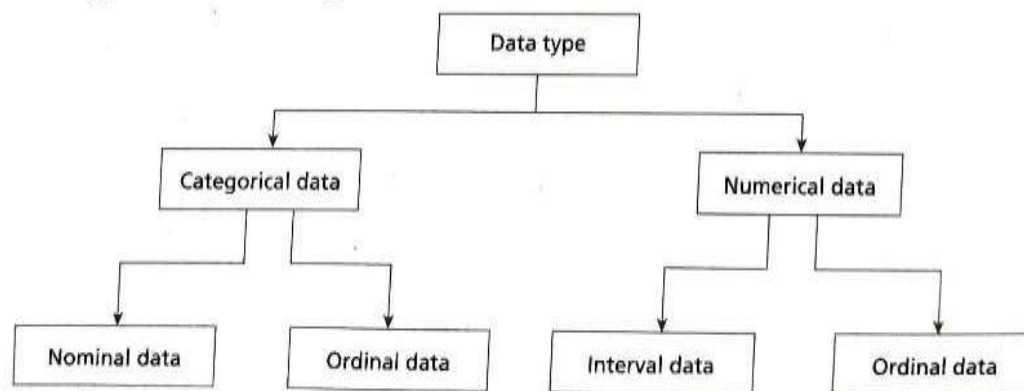


Figure 2.1: Types of Data

The data types are shown in Figure 2.1.

Categorical or Qualitative data:

The categorical data can be divided into two types. They are

Nominal Data - In Table 2.2, patient ID is nominal data. Nominal data are symbols and cannot be processed like a number. For example, the average of a patient ID does not make any statistical sense. Nominal data type provides only information but has no ordering among data. Only operations like (-,) are meaningful for these data. For example, the patient ID can be checked for equality and nothing else.

Ordinal Data - It provides enough information and has natural order. For example, Fever = {Low, Medium, High} is an ordinal data. Certainly, low is less than medium and medium is less than high, irrespective of the value. Any transformation can be applied to these data to get a new value.

Numeric or Quantitative Data: It can be divided into two categories.

They are interval type and ratio type.

Interval Data - Interval data is numeric data for which the differences between values are meaningful. For example, there is a difference between 30 degrees and 40 degrees. Only the permissible operations are + and -.

Ratio Data - For ratio data, both differences and ratio are meaningful. The difference between the and interval data is the position of zero in the scale. For example, take the Centigrade-Fahrenheit conversion. The zeroes of both scales do not match.

Another way of classifying the data is to classify it as:

1. Discrete value data
2. Continuous data

Discrete Data : This kind of data is recorded as integers.

For example, the responses of the survey can be discrete data. Employee identification number such as 10001 are discrete data.

Continuous Data: It can be fitted into a range and includes decimal point. For example, age is a continuous data. Though age appears to be discrete data, one may be 12.5 years old and it makes sense. Patient height and weight are all continuous data.

Third way of classifying the data is based on the number of variables used in the dataset.

1. Univariate data: Dataset has one variable.
2. Bivariate data : Dataset has two variables.
3. Multivariate data: Dataset has multivariable uses three or more variables

UNIVARIATE DATA ANALYSIS AND VISUALIZATION

Data Visualization

To understand data, graph visualization is a must. Data visualization helps to understand data. It helps to present information and data to customers.

Some of the graphs that are used in univariate data analysis are bar charts, histograms, frequency polygons and pie charts.

The advantages of the graphs are presentation of data, summarization of data, description of data, exploration of data, and to make comparisons of data. Let us consider some forms of graphs now:

Bar Chart: A Bar chart (or Bar graph) is used to display the frequency distribution for variables. Bar charts are used to illustrate discrete data. The charts can also help to explain the counts of nominal data. It also helps in comparing the frequency of different groups.

The bar chart for students' marks (45, 60, 60, 80, 85) with Student ID = {1, 2, 3, 4, 5} is shown below in Figure 2.3.

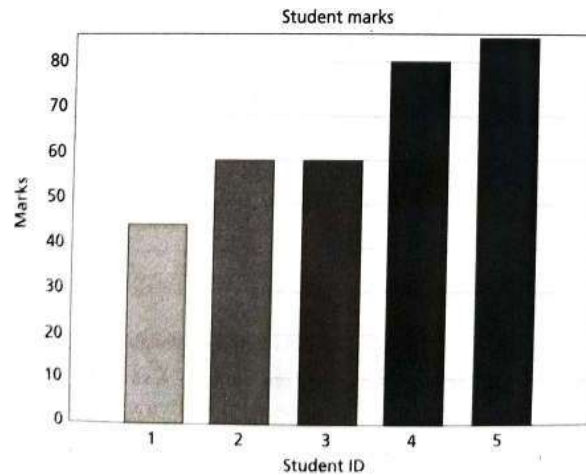
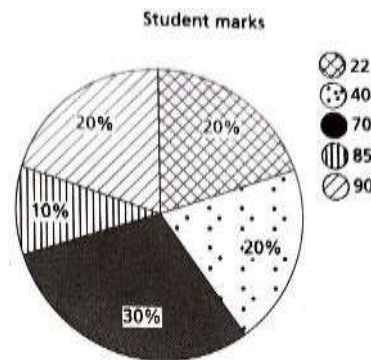


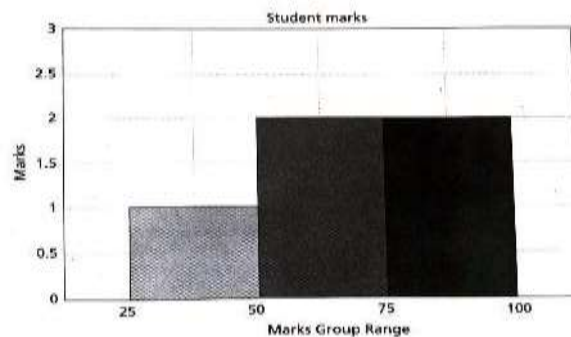
Figure 2.3: Bar Chart

Pie Chart: These are equally helpful in distribution of students' illustrating the univariate data. The percentage frequency marks {22, 22, 40, 40, 70, 70, 70, 85, 90, 90} is below in Figure 2.4.



It can be observed that the number of students with 22 marks are 2. The total number of students are 10. So, $2/10 \times 100 = 20\%$ space in a pie of 100% is allotted for marks 22 in Figure above

Histogram: It plays an important role in data mining for showing frequency distributions. The histogram for students' marks {45, 60, 60, 80, 85} in the group range of 0-25, 26-50, 51-75, 76-100 is given below in Figure 2.5. One can visually inspect from Figure 2.5 that the number of students in the range 76-100 is 2.



Histogram conveys useful information like the nature of data and its mode. Mode indicates the peak of the dataset. In other words, histograms can be used as charts to show frequency, skewness present in the data, and shape.

Dot Plots: These are similar to bar charts. They are less clustered as compared to bar charts as they illustrate the bars only with single points. The dot plot of English marks for five students with ID as {1,2,3,4,5} and marks {45,60,60,80,85} is given fig 2.6 below. The advantage is that by visual inspection one can find out who got more marks.

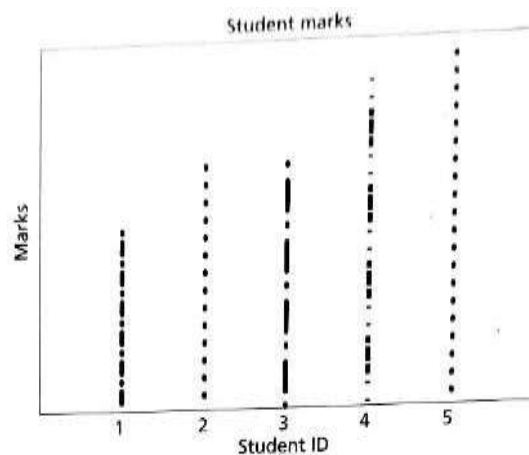


Figure 2.6: Dot Plots

Central Tendency

One cannot remember all the data. Therefore, a condensation or summary of the data is necessary. This makes the data analysis easy and simple. One such summary is called central tendency. Thus, central tendency can explain the characteristics of data and that further helps in comparison. Mass data have a tendency to concentrate at certain values, normally in the central location. It is called the measure of central tendency (or averages). This represents the first order of measures.

Popular Measures are mean, median and mode.

1. **Mean** - Arithmetic average (or mean) is a measure of central tendency that represents the center of the dataset. This is the commonest measure used in our daily conversation such as average income or average traffic. It can be found by adding all the data and dividing the sum by the number of observations.

Let $x, x_1, x_2, x_3, x_4, \dots, x_n$ be a set of 'N' values or observations, then the arithmetic mean is given as:

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_N}{N} = \frac{1}{N} \sum_{i=1}^N x_i$$

Weighted mean - Unlike arithmetic mean that gives the weightage of all items equally weighted mean gives different importance to all items as the item importance varies. Hence, different weightage can be given to items.

In case of frequency distribution, mid values of the range are taken for This is computation. illustrated in the following computation.

In weighted mean, the mean is computed by adding the product of proportion and group mean. It is mostly used when the sample sizes are unequal.

Geometric mean -Let $x_1, x_2, x_3, x_4 \dots x_n$ be a set of 'N' values or observations. Geometric mean is the Nth root of the product of N items. The formula for computing, geometric mean is given as follows

$$\text{Geometric mean} = \left(\prod_{i=1}^n x_i \right)^{\frac{1}{N}} = \sqrt[N]{x_1 \times x_2 \times \dots \times x_N}$$

2. **Median** : The middle value in the distribution is called median. If the total number of items in the distribution is odd, then the middle value is called 'median. If the numbers are even, then the average value of two items in the center is the median. It can be observed that the median is the value where x is divided into two equal halves, with half of the values being lower than the median and half higher than the median. A median class is that class where (N/2)th item is present.

3. **Mode**: Mode is the value that occurs more frequently in the dataset. In other words, the value that has the highest frequency is called mode. Mode is only for discrete data and is not applicable for continuous data as there are no repeated values in continuous data.

The procedure for finding the mode is to calculate the frequencies for all the values in the data, and mode is the value (or values) with the highest frequency. Normally, the dataset is classified as unimodal, bimodal and trimodal with modes 1, 2 and 3, respectively.

Dispersion

The spread out of a set of data around the central tendency (mean, median or mode) is called dispersion. Dispersion is represented by various ways such as range, variance, standard deviation and standard error, These are second order measures.

The most common measures of the dispersion data are listed below:

Range: Range is the difference between the maximum and minimum of values of the given lies of data.

Standard Deviation: The mean does not convey much more than a middle point. For example, the following datasets {10, 20, 30} and {10, 50, 0} both have a mean of 20. The difference between these two sets is the spread of data. Standard deviation is the average distance from the mean of the dataset to each point

The formula for sample standard deviation is given by:

$$\sigma = \sqrt{\frac{\sum_{i=1}^N (x_i - \bar{x})^2}{N - 1}}$$

Quartiles and InterQuartile Range: It is sometimes convenient to subdivide the dataset using coordinates. Percentiles are about data that are less than the coordinates by some percentage of the total value. k^{th} percentile is the property that the $k\%$ of the data lies at or below X_i , For example, median is 50^{th} percentile and can be denoted as $Q_{0.50}$ The 25^{th} percentile is called first quartile (Q_1) and the 75^{th} percentile is called third quartile (Q_3).

Another measure that is useful to measure dispersion is InterQuartile Range (IQR). The IQR is the difference between Q_1 & Q_3

Interquartile percentile = $Q_1 - Q_3$

Outliers are normally the values falling apart at least by the amount $1.5 * \text{IQR}$ above the third quartile or below the first quartile

Interquartile is defined by $Q_{0.75} - Q_{0.25}$.

Five-point Summary and BoX Plots: The median, quartiles, and , and maximum minimum written in the. order < Minimum, O., Median, O., Maximum > is known as a five-point summary.

Box plots are suitable for continuous variables and a nominal variable. Box plots can be used to illustrate data distributions and summary of data. It is the popular way for plotting five number summaries. A BoX plot is also known as a Box and whisker plot.

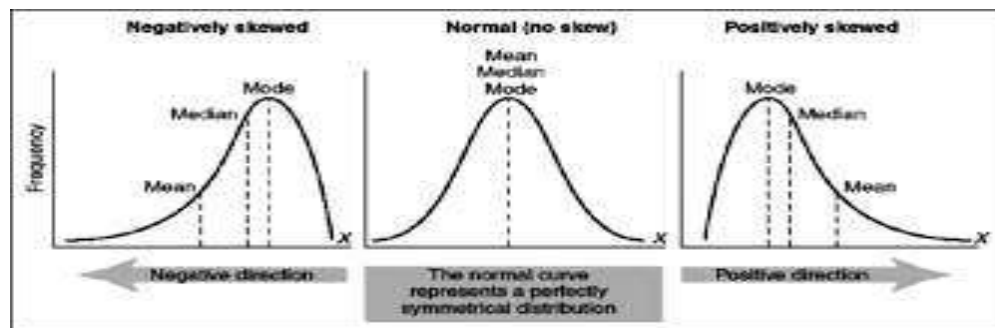
The box Contains bulk of the data. These data are between the first and third quartiles. The line inside the box indicates location - mostly median of the data. If the median is not equidistant, then the data is skewed. The whiskers that project from the ends of the box indicate the spread of the tails and the maximum and minimum of the data value.

Shape

Skewness and Kurtosis (called moments) indicate the symmetry/asymmetry and peak location of the dataset.

Skewness

The measures of direction and degree of symmetry are called measures of third order. Ideally, Skewness should be zero as in ideal normal distribution. More often, the given dataset may not have perfect symmetry (consider the following Figure 2.8).



The dataset may also either have very high values or extremely low values. If the dataset has far higher values, then it is said to be skewed to the right. On the other hand, if the dataset has far more low values then it is said to be skewed towards the left. If the tail is longer on the left-hand side and hump on the right-hand side, it is called positive skew. Otherwise, it is called negative skew.

The given dataset may have an equal distribution of data. The implication of this is that if the data is skewed, then there is a greater chance of outliers in the dataset. This affects the mean and median. Hence, this may affect the performance of the data mining algorithm. A perfect symmetry means the skewness is zero. In the case of skew, the median is greater than the mean. In positive skew, the mean is greater than the median.

Generally, for negatively skewed distribution, the median is more than the mean. The following measure is more commonly used to measure skewness. Let $X_1, X_2, \dots, X_N$ be a set of 'N' values or observations then the skewness can be given as:

$$\frac{1}{N} \times \sum_{i=1}^N \frac{(x_i - \mu)^3}{\sigma^3}$$

Here, μ is the population mean and σ is the population standard deviation of the univariate data. Sometimes, for bias correction instead of N, N - 1 is used.

Kurtosis

Kurtosis also indicates the peaks of data. If the data is high peak, then it indicates higher kurtosis and vice versa.

Kurtosis is the measure of whether the data is heavy tailed or light tailed relative to normal distribution. It can be observed that normal distribution has a bell-shaped curve with no long tails. Low Kurtosis tends to have light tails. The implication is that there is no outlier data. Let $x_1, x_2, \dots, x_N$ be a set of 'N' values or observations. Then, kurtosis is measured using the formula given below:

$$\frac{\sum_{i=1}^N (x_i - \bar{x})^4 / N}{\sigma^4}$$

Some of the other useful measures for finding the shape of the univariate dataset are mean absolute deviation (MAD) and coefficient of variation (CV).

Mean Absolute Deviation (MAD)

MAD is another dispersion measure and is robust to outliers. Normally, the outlier point is detected by computing the deviation from median and by dividing it by MAD. Here, the absolute deviation between the data and mean is taken. Therefore, the mean absolute deviation is given as:

$$\frac{\sum |x - \mu|}{N}$$

Coefficient of Variation (CV)

Coefficient of variation is used to compare datasets with different units. CV is the ratio of standard deviation and mean, and %CV is the percentage of coefficient of variations.

Special Univariate Plots

The ideal way to check the shape of the dataset is a stem and leaf plot. A stem and leaf plot are a display that helps us to know the shape and distribution of the data. In this method, each value is split into a 'stem' and a

leaf. The last digit is usually the leaf and digits to the left of the leaf mostly form the stem. For example, marks 45 are divided into stem 4 and leaf 5 in Figure 2.9.

The stem and leaf plot for the English subject marks, say, {45, 60, 60, 80, 85} is given in Figure 2.9.

Stem	Leaf
4	5
5	
6	0 0
7	
8	0 5

Figure 2.9: Stem and Leaf Plot for English Marks

It can be seen from Figure 2.9 that the first column is stem and the second column is leaf. For the given English marks, two students with 60 marks are shown in stem and leaf plot as stem-6 with 2 leaves with 0.

As discussed earlier, the ideal shape of the dataset is a bell-shaped curve. This corresponds to normality. Most of the statistical tests are designed only for normal distribution of data.

A Q-Q plot can be used to assess the shape of the dataset. The Q-Q plot is a 2D scatter plot of an univariate data against theoretical normal distribution data or of two datasets - the quantiles of the first and second datasets. The normal Q-Q plot for marks = [13 11 23 48 9] is given below in Figure 2.10.

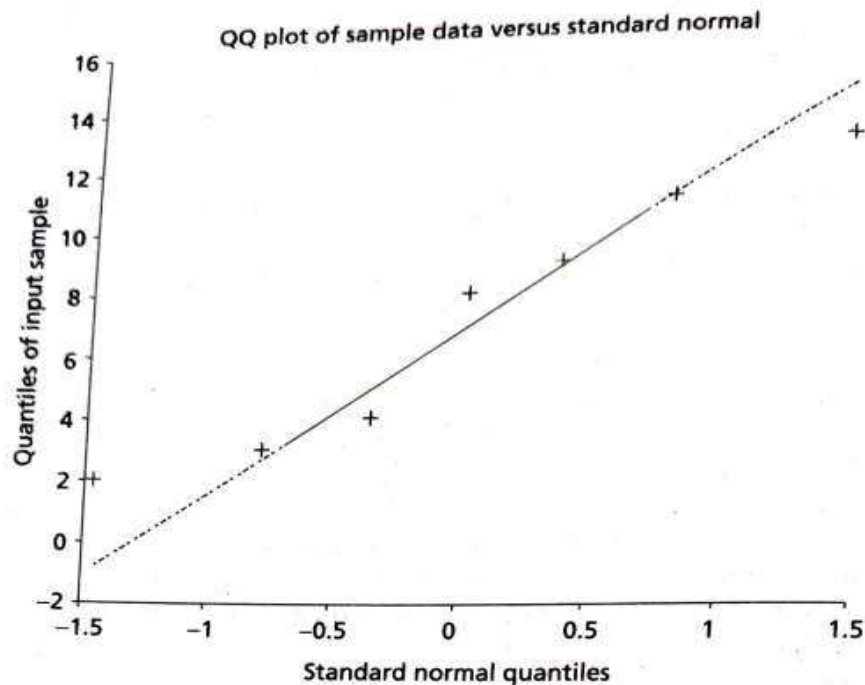


Figure 2.10: Normal Q-Q Plot

ideally, the points fall along the reference line (45 Degree) if the data follows normal distribution. If the deviation is more, then there is greater evidence that the datasets follow some different distribution, that is, other than the normal distribution shape. In such a case, careful analysis of the statistical investigations should be carried out before interpretation.

This skewness, kurtosis, mean absolute deviation and coefficient of variation help in assessing the univariate data.

BIVARIATE DATA AND MULTIVARIATE DATA

Bivariate Data involves two variables. Bivariate data deals with causes of relationships. The aim is to find relationships among data. Consider the following Table 2.3, with data of the temperature in a shop and sales of sweaters.

Table 2.3: Temperature in a Shop and Sales Data

Temperature (in centigrade)	Sales of Sweaters (in thousands)
5	200
10	150
15	140
20	75
22	60
23	55
25	20

Here, the aim of bivariate analysis is to find relationships among variables. The relationships can then be used in comparisons, finding causes, and in further explorations. To do that, graphical display of the data is necessary. One such graph method is called scatter plot.

Scatter plot is used to visualize bivariate data. It is useful to plot two variables with or without nominal variables, to illustrate the trends, and also to show differences. It is a plot between explanatory and response variables. It is a 2D graph showing the relationship between two variables.

The scatter plot (Refer Figure 2.11) indicates strength, shape, direction and the presence of Outliers. It is useful in exploratory data before calculating a correlation coefficient or fitting regression curve.

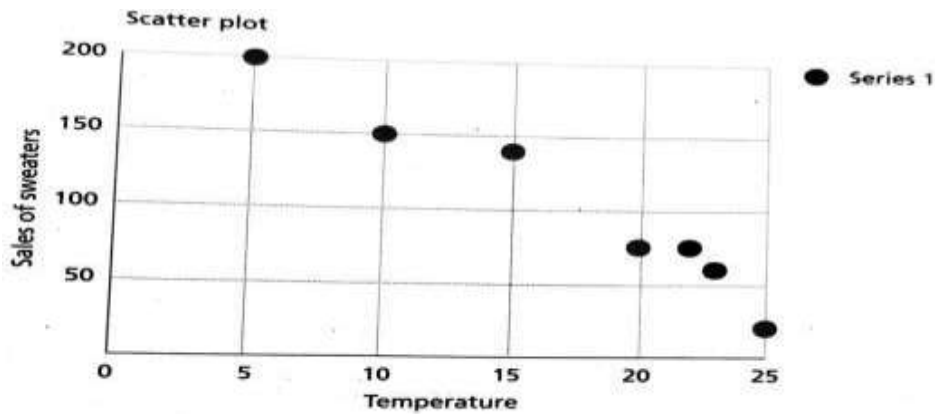


Figure 2.11: Scatter Plot

Line graphs are similar to scatter plots. The Line Chart for sales data is shown in Figure 2.12.

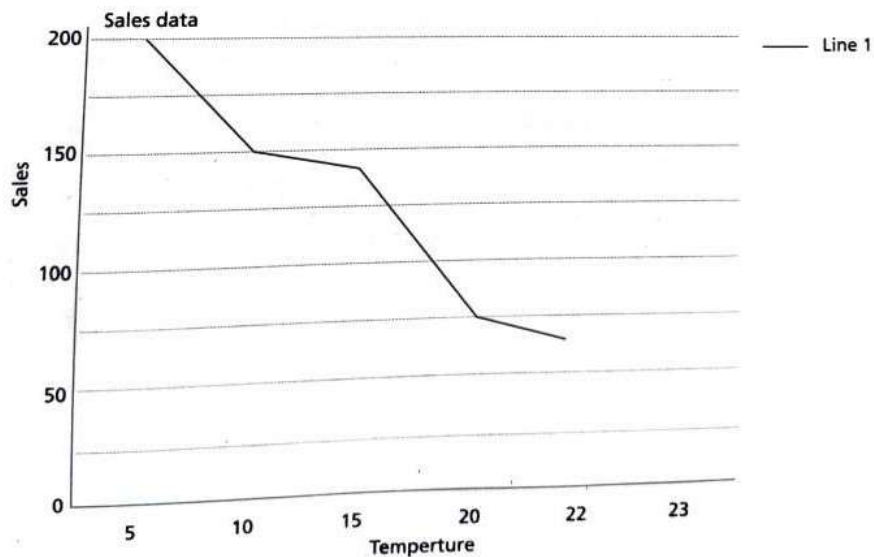


Figure 2.12: Line Chart

Bivariate Statistics

Covariance and Correlation are examples of bivariate statistics. Covariance is a measure of joint probability of random variables, say X and Y . Generally, random variables are represented in capital letters. It is defined as $\text{covariance}(X, Y)$ or $\text{COV}(X, Y)$ and is used to measure the variance between two dimensions. The formula for finding covariance for specific x , and y are:

$$\text{cov}(X, Y) = \frac{1}{N} \sum_{i=1}^N (x_i - E(X))(y_i - E(Y))$$

Here, x_i and y_i , are data values from X and Y . $E(X)$ and $E(Y)$ are the mean values of x_i and y_i . N is the number of given data. Also, the $\text{COV}(X, Y)$ is same as $\text{COV}(Y, X)$.

Correlation

The Pearson correlation coefficient is the most common test for determining any association between two phenomena. It measures the strength and direction of a linear relationship between the x and y variables.

The correlation indicates the relationship between dimensions using its sign. The sign is more important than the actual value.

1. If the value is positive, it indicates that the dimensions increase together.
2. If the value is negative, it indicates that while one dimension increases, the other dimension decreases.
3. If the value is zero, then it indicates that both the dimensions are independent of each other.

If the dimensions are correlated, then it is better to remove one dimension as it is a redundant dimension.

If the given attributes are $X = (x_1, x_2, \dots, x_N)$ and $Y = (y_1, y_2, \dots, y_N)$, then the Pearson correlation coefficient, that is denoted as r , is given as:

$$r = \frac{\text{COV}(X, Y)}{\sigma_X \sigma_Y}$$

MULTIVARIATE STATISTICS

In machine learning, almost all datasets are multivariable. Multivariate data is the analysis of more than two observable variables, and often, thousands of multiple measurements need to be conducted for one or more subjects.

The multivariate data is like bivariate data but may have more than two dependant variables. Some of the multivariate analysis are regression analysis, principal component analysis, and path analysis.

<i>Id</i>	<i>Attribute 1</i>	<i>Attribute 2</i>	<i>Attribute 3</i>
1	1	4	1
2	2	5	2
3	3	6	1

Multivariate data has three or more variables. The aim of the multivariate analysis is much more. They are regression analysis, factor analysis and multivariate analysis of variance.

Heatmap

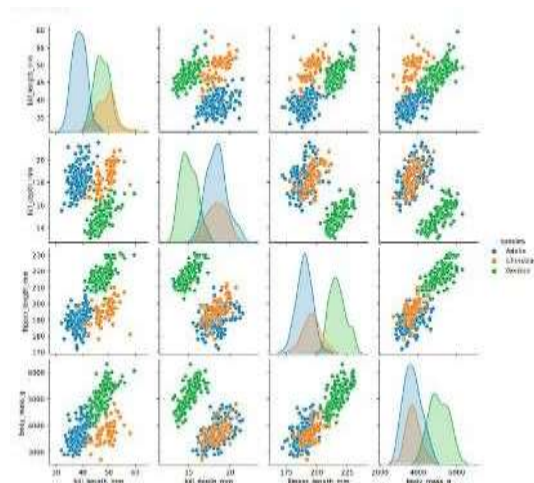
Heatmap is a graphical representation of 2D matrix. It takes a matrix as input and colours it. The darker colours indicate very large values and lighter colors indicate smaller values. The advantage of this method is that humans perceive colours well. So, by colour shaping, larger values can be perceived well. For example, in vehicle traffic data, heavy traffic regions can be differentiated from low traffic regions



through heatmap.

Pair Plot

Pairplot or scatter matrix is a data visual technique for multivariate data. A scatter matrix consists of several pairwise scatter plots of variables of multivariate data. All the results are presented in a matrix format . By visual examination of the chart , one can easily find relationships among the variables such as correlation between variables.



ESSENTIAL MATHEMATICS FOR MULTIVARIATE DATA

Machine learning involves many mathematical concepts from the domain of Linear algebra, Statistics, Probability and Information theory. The subsequent sections discuss important aspects of linear algebra and probability.

Linear Systems and Gaussian Elimination for Multivariate Data

A linear system of equations is a group of equations with unknown variables. Let $Ax = y$, then the solution x is given as:

$$x = y/A = A^{-1} y \quad (2.19)$$

This is true if y is not zero and A is not zero. The logic can be extended for N-set of equations with 'n' unknown variables.

It means if $A =$ and $y = (y_1 y_2 \dots y_n)$, then the unknown variable x can be computed as:

$$x = y/A = A^{-1} y \quad (2.20)$$

If there is a unique solution, then the system is called consistent independent. If there are various solutions, then the system is called consistent dependant. If there are no solutions and if the equations are contradictory, then the system is called inconsistent.

For solving large number of system of equations, Gaussian elimination can be used. The procedure for applying Gaussian elimination is given as follows:

1. Write the given matrix.
2. Append vector y to the matrix A . This matrix is called augmentation matrix.
3. Keep the element a_{11} as pivot and eliminate all a_{11} in second row using the matrix operation,

$$R_2 - \left(\frac{a_{21}}{a_{11}} \right), \text{ here } R_2 \text{ is the second row and } \left(\frac{a_{21}}{a_{11}} \right) \text{ is called the multiplier.}$$

The same logic can be used to remove a_{11} in all other equations.

4. Repeat the same logic and reduce it to reduced echelon form. Then, the unknown variable as:

$$x_s = \frac{y_{ns}}{a_{ns}} \quad (2.21)$$

5. Then, the remaining unknown variables can be found by back-substitution as:

$$x_{s-1} = \frac{y_{s-1} - a_{s-1} \times x_s}{a_{(s-1)(s-1)}} \quad (2.22)$$

This part is called backward substitution.

To facilitate the application of Gaussian elimination method, the following row operations are applied:

1. Swapping the rows
2. Multiplying or dividing a row by a constant
3. Replacing a row by adding or subtracting a multiple of another row to it

Matrix Decomposition

It is often necessary to reduce a matrix to its constituent parts so that complex matrix operations can be performed.

Then, the matrix A can be decomposed as:

$$A = Q \Lambda Q^T \quad (2.23)$$

matrix of eigen vectors, Λ is the diagonal matrix and Q^T is the transpose of matrix Q .

LU Decomposition

One of the simplest matrix decomposition is LU decomposition where the matrix A can be decomposed matrices: $A = LU$

Here, L is the lower triangular matrix and U is the upper triangular matrix. The decomposition can be done using Gaussian elimination method as discussed in the previous section. First, an identity matrix is augmented to the given matrix. Then, row operations and Gaussian elimination is applied to reduce the given matrix to get matrices L and U.

Machine Learning and Importance of Probability and Statistics

Machine learning is linked with statistics and probability. Like linear algebra, statistics is the heart of machine learning. The importance of statistics needs to be stressed as without statistics;

Probability Distributions

A probability distribution of a variable, say X, summarizes the probability associated with X's events. Distribution is a parameterized mathematical function. In other words, distribution is a function that describes the relationship between the observations in a sample space. Consider a set of data. The data is said to follow a distribution if it obeys a mathematical function that characterizes that distribution. The function can be used to calculate the probability of individual observations.

Probability distributions are of two types:

1. Discrete probability distribution
2. Continuous probability distribution

The relationships between the events for a continuous random variable and their probabilities

Continuous Probability Distributions Normal, Rectangular, and Exponential distributions fall under this category.

1. **Normal Distribution** – Normal distribution is a continuous probability distribution. This is also known as gaussian distribution or bell-shaped curve distribution. It is the most common distribution function. The shape of this distribution is a typical bell-shaped curve. In normal distribution, data tends to be around a central value with no bias on left or right. The heights of the students, blood pressure of a population, and marks scored in a class can be approximated using normal distribution. PDF of the normal distribution is given as:

$$f(x, \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad (2.24)$$

Here, μ is mean and σ is the standard deviation. Normal distribution is characterized by two parameters – mean and variance.

One important concept associated with normal distribution is z-score. It can be computed as:

$$z = \frac{x - \mu}{\sigma}. \text{ When } \mu \text{ is zero and } \sigma \text{ is 1, z-score is same as } x.$$

This is useful to normalize the data.

2. **Rectangular Distribution** – This is also known as uniform distribution. It has equal probabilities for all values in the range a, b. The uniform distribution is given as follows:

$$P(X = x) = \begin{cases} \frac{1}{b-a} & \text{for } a \leq x \leq b \\ 0 & \text{Otherwise} \end{cases} \quad (2.25)$$

3. **Exponential Distribution** – This is a continuous uniform distribution. This probability distribution is used to describe the time between events in a Poisson process. Exponential distribution is another special case of Gamma distribution with a fixed parameter of 1. This distribution is helpful in modelling of time until an event occurs.

The PDF is given as follows:

$$f(x, \lambda) = \begin{cases} \lambda e^{-\lambda x} & \text{if } x \geq 0 \\ 0 & \text{if } x < 0 \end{cases} \quad (\lambda > 0) \quad (2.26)$$

Here, x is a random variable and λ is called rate parameter. The mean and standard deviation of exponential distribution is given as β , where, $\beta = \frac{1}{\lambda}$.

Discrete Distribution Binomial, Poisson, and Bernoulli distributions fall under this category.

1. **Binomial Distribution** – Binomial distribution is another distribution that is often encountered in machine learning. It has only two outcomes: success or failure. This is also called Bernoulli trial.

The objective of this distribution is to find probability of getting success k out of n trials. The way to get success out of k out of n number of trials is given as:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!} \quad (2.27)$$

The binomial distribution function is given as follows, where p is the probability of success and probability of failure is $(1 - p)$. The probability of success in a certain number of trials is given as:

$$p^k(1-p)^{n-k} \text{ or } p^k q^{n-k} \quad (2.28)$$

Combining both, one gets PDF of binomial distribution as:

$$\binom{n}{k} p^k (1-p)^{n-k} \quad (2.29)$$

Here, p is the probability of each choice, k is the number of choices, and n is the total

$$\mu = n \times p \quad (2.30)$$

number of choices. The mean of binomial distribution is given below:

$$\sigma^2 = np(1-p) \quad (2.31)$$

Hence, the standard deviation is given as

$$\sigma = \sqrt{np(1-p)} \quad (2.32)$$

2. Poisson Distribution – It is another important distribution that is quite useful. Given an interval of time, this distribution is used to model the probability of a given number of events k . The mean rule is inclusive of previous events. Some of the examples of Poisson distribution are number of emails received, number of customers visiting a shop and the number of phone calls received by the office.

The PDF of Poisson distribution is given as follows:

$$f(X = x; \lambda) = \text{Pr}[X = x] = \frac{e^{-\lambda} \lambda^x}{x!} \quad (2.33)$$

Here, x is the number of times the event occurs and λ is the mean number of times an event occurs.

The mean is the population mean at number of emails received and the standard deviation is $\sqrt{\lambda}$.

3. Bernoulli Distribution – This distribution models an experiment whose outcome is binary. The outcome is positive with p and negative with $1 - p$. The PMF of this distribution is given as:

$$f(k; p) = \begin{cases} q = 1 - p & \text{if } k = 0 \\ p & \text{if } k = 1. \end{cases} \quad (2.34)$$

The mean is p and variance is $p(1 - p) = q$

Density Estimation

Let there be a set of observed values $x_1, x_2, \dots, x_n$ from a larger set of data whose distribution is not known. Density estimation is the problem of estimating the density function from an observed data. There are two types of density estimation methods, namely parametric density estimation and non-parametric density estimation.

Parametric Density Estimation It assumes that the data is from a known probabilistic distribution and can be estimated as $p(x | \theta)$, where, θ is the parameter. Maximum likelihood function is a parametric estimation method.

Maximum Likelihood Estimation For a sample of observations, one can estimate the probability distribution. This is called density estimation. Maximum Likelihood Estimation (MLE) is a probabilistic framework that can be used for density estimation. This involves formulating a function called likelihood function which is the conditional probability of observing the observed samples and distribution function with its parameters. For example, if the observations are $X = \{x_1, x_2, \dots, x_n\}$, then density estimation is the problem of choosing a PDF with suitable parameters to describe the data. MLE treats this problem as a search or optimization problem where the probability should be maximized for the joint probabilities of X and its parameter, theta.

For example, this is expressed as $p(X; \theta)$, where, $X = [x_1, x_2, \dots, x_n]$

The likelihood of observing the data is given as a function $L(X; \theta)$. The objective of MLE is to maximize this function as $\max L(X; \theta)$.

The joint probability of this problem can be restated as $\prod_{i=1}^n p(x_i; \theta)$.

The computation of the above formula is unstable and hence the problem is restated as maximum of log conditional probability given θ . This is given as:

$$\sum_{i=1}^n \log p(x_i; \theta) \quad (2.35)$$

Instead of maximizing, one can minimize this function as:

$\min = -\sum_{i=1}^n \log p(x_i; \theta)$ as, often, minimization is preferred over maximization.

This is called negative log-likelihood function.

If one assumes that the regression problem can be framed as predicting output y given input x , then for $p(y/x)$, the MLE framework can be applied as:

$$\max \sum \log(y | x_i, h) \quad (2.36)$$

Here, h is the linear regression model. If Gaussian distribution is assumed as it is an obvious fact that most of the data follow Gaussian distribution, then MLE can be stated as:

$$\max \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y_i - h(x_i, \theta))^2}{2\sigma^2}} \quad (2.37)$$

Here, b is the regression coefficient and x_i is the given sample. One can maximize this function or minimize the negative log likelihood function to provide a solution for linear regression problem. The Eq. (2.37) yields the same answer of the least-square approach.

Gaussian Mixture Model and Expectation-Maximization (EM) Algorithm In machine learning, clustering is one of the important tasks. It is discussed in Chapter 13. MLE framework is quite useful for designing model-based methods for clustering data. A model is a statistical method and data is assumed to be generated by a distribution model with its parameter, theta. There may be many distributions involved and that is why it is called as mixture model.

Generally, there can be many unspecified distributions with different set of parameters. The EM algorithm has two stages:

1. Expectation (E) Stage – In this stage, the expected PDF and its parameters are estimated for each latent variable.
2. Maximization (M) stage – In this, the parameters are optimized using the MLE function. This process is iterative, and the iteration is continued till all the latent variables are fitted by probability distributions effectively along with the parameters.

Non-parametric Density Estimation A non-parametric estimation can be generative or discriminative. Parzen window is a generative estimation method that finds $p(x | \Theta)$ as conditional density.

Discriminative

methods directly compute $p(x | \Theta)$ as posteriori probability. Parzen window and k-Nearest Neighbour (KNN) rule are examples of non-parametric density estimation.

Parzen Window Let there be 'n' samples, $X = \{x_1, x_2, \dots, x_n\}$

The samples are drawn independently, called as identically independent distribution. Let R be the region that covers 'k' samples of total 'n' samples. Then, the probability density function is given as:

$$p = k/n \quad (2.38)$$

The estimate is given as:

$$p(x) = \frac{k/n}{V} \quad (2.39)$$

where, V is the volume of the region R. If R is the hypercube centered at x and h is the length of the hypercube, the volume V is h^2 for 2D square cube and h^3 for 3D cube.

The Parzen window is given as follows:

$$\varphi\left(\frac{x_i - x}{h}\right) = \begin{cases} 1 & \text{if } \frac{|x_i - x|}{h} < \frac{1}{2} \\ 0 & \text{otherwise} \end{cases} \quad (2.40)$$

The window indicates if the sample is inside the region or not. The Parzen probability density function estimate using Eq. (2.40) is given as:

$$\begin{aligned} p(x) &= \frac{k/n}{V} \\ &= \frac{1}{n} \sum_{i=1}^n \frac{1}{V_n} \varphi\left(\frac{x_i - x}{h}\right) \end{aligned} \quad (2.41)$$

This window can be replaced by any other function too. If Gaussian function is used, then it is called Gaussian density function.

KNN Estimation The KNN estimation is another non-parametric density estimation method. Here, the initial parameter k is determined and based on that k-neighbours are determined. The probability density function estimate is the average of the values that are returned by the neighbours.

OVERVIEW OF HYPOTHESIS

Data collection alone is not enough. Data must be interpreted to give a conclusion. The conclusion should be a structured outcome. This assumption of the outcome is called a hypothesis. Statistical methods are used to confirm or reject the hypothesis. The assumption of the statistical test is called null hypothesis. It is also called as hypothesis zero (H_0). In other words, hypothesis is the existing belief. The violation of this hypothesis is called first hypothesis (H_1) or hypothesis one.

This is the hypothesis the researcher is trying to establish.

There are two types of hypothesis tests, parametric and non-parametric. Parametric tests are based on parameters such as mean and standard deviation. Non-parametric tests are dependent on characteristics such as independence of events or data following certain distribution. Statistical tests help to:

1. Define null and alternate hypothesis
2. Describe the hypothesis using parameters
3. Identify the statistical test and statistics
4. Decide the criteria called significance value α
5. Compute p-value (probability value)
6. Take the final decision of accepting or rejecting the hypothesis based on the parameters

Hypothesis testing is particularly important as it is an integral part of the learning algorithms. Generally, the data size is small. So, one may have to know whether the hypothesis will work for additional samples and how accurate it is. No matter how effective the statistical tests are, two kinds of errors are involved, that are Type I and Type II.

Type I error is the incorrect rejection of a true null hypothesis and is called false positive.

Type II error is the incomplete failure of rejecting a false hypothesis and is called false negative.

During these calculations, one must include the size of the data sample. Degree of freedom indicates the number of independent pieces of information used for the test. It is indicated as n . The mean or variance can be used to indicate the degree of freedom.

Hypothesis Testing

Let us define two important errors called sample error and true (or actual) error. Let us assume that D is the unknown distribution, Target function is $f(x)$: $x \in \{0, 1\}$, x is the instance, $h(x)$ is the hypothesis, and sample set is S that derives the samples on instances drawn from X . Then, the actual error is denoted as:

$$\text{error}_D(h) = \Pr_{x \in D} \{f(x) \neq h(x)\} \quad (2.42)$$

In other words, true error is the probability that the hypothesis will misclassify an instance that is drawn at random. The point is that population is very large and hence it is not possible to determine true error and can only be estimated. So, another error is called sample error or estimator.

Sample error is with respect to sample S . It is the probability for instances drawn from X , that is,

$$\text{error}_S(h) = \frac{1}{n} \sum_{x \in S} \delta(f(x), h(x)) \quad (2.43)$$

Here, $\delta(f(x), h(x))$ is 1 if $f(x) \neq h(x)$, otherwise is zero.

the fractions of S that are misclassified. The sample error is given as follows:

p-value

Statistical tests can be performed to either accept or reject the null hypothesis. This is done by the value called p-value or probability value. It indicates the probability of hypothesis being true. The p-value is used to interpret or quantify the test. For example, a statistical test result may give a value of 0.03. Then, one can compare it with the level 0.05. As $0.03 < 0.05$, the result is assumed to be significant. This means that the variables tested are not independent. Here, 0.05 is called significant level. In general, significant level is called α and p-value is compared with α . If $p\text{-value} \leq \alpha$, then the hypothesis H_1 is rejected and if $p\text{-value} > \alpha$, then the hypothesis H_0 is rejected.

Confidence Intervals

The acceptance or rejection of the hypothesis can also be done using confidence interval.

The confidence interval is computed as:

$$\text{Confidence interval} = 1 - \text{significant level} \quad (2.44)$$

Confidence level is the range of values that indicates the location of true mean. Confidence intervals indicate the confidence of the result. If the confidence level is 90%, then it infers that there is 90% of chance that the true mean lies in this range and remaining 10% indicates that true mean is not present. For finding this, one requires mean and standard deviation. Then, $\bar{x}$ can be given as

$$\text{mean}(\bar{x}) \pm z \times \frac{s}{\sqrt{N}}$$

Here, s is the standard deviation, N is the number of samples, and z is the value associated with 90% and is called % of confidence. This is also called as margin of error. Sample error is the unbiased estimate of true error. If no information is provided, then both errors are the same. It is, however, often safe to suggest a margin of confidence associated with the hypothesis. The hypothesis with 95% confidence about the sample error can be given as follows:

$$\text{error}_s(h) \pm 1.96 \sqrt{\frac{\text{error}_s(h)(1 - \text{error}_s(h))}{n}} \quad (2.45)$$

This 1.96 indicates the 95% confidence of the error. The number 1.96 can be replaced by any number that is associated with different levels of confidence.

The procedure to estimate the difference between two hypothesis, say h_1 and h_2 , is as follows:

1. A parameter d can be chosen to estimate the error of two hypothesis:

2. $d \equiv \text{error}_D(h_1) - \text{error}_D(h_2)$ (2.46)

Here, there are two hypothesis h_1 and h_2 tested on two sample sets s_1 and s_2 . Similarly, n_1 and n_2 are randomly drawn number of samples.

3. The estimator $\hat{d}$ can be estimated as the difference as:

$$\hat{d} \equiv \text{error}_D(h_1) - \text{error}_D(h_2)$$

4. The confidence intervals can be used for the estimator also as follows:

$$\hat{d} \pm z \sqrt{\frac{\text{error}_{s_1}(h_1) - (1 - \text{error}_{s_1}(h_1))}{n_1} + \frac{\text{error}_{s_2}(h_2) - (1 - \text{error}_{s_2}(h_2))}{n_2}} \quad (2.47)$$

Sometimes, it is desirable to find interval L and U such that $N\%$ of the probability falls in this

Comparing Learning Methods

Some of the methods for comparing the learning programs are given below:

Z-test

Z-test assumes normal distribution of data whose population variation is known

$$Z = \frac{X - \mu}{\sqrt{\frac{\sigma^2}{n}}} \quad (2.48)$$

Here, X is the input data, and n is number of data elements. μ and σ are mean and standard deviation of X , respectively.

t-test and Paired t-test

t-test is a hypothesis test and checks if the difference between two samples' mean is real or by chance. Here, data is continuous and randomly selected. There will only be small number of samples and variance between groups is real. The t-test statistics follows t-distribution under null hypothesis and is used when the number of samples

<30. So, the procedure is:

- Select a group
- Compute average

Compute it with theoretical value and compute t-statistic:

$$t = \frac{m - \mu}{\frac{s}{\sqrt{n}}} \quad (2.49)$$

Here, t is t-statistic, m is the mean of the group, m is the theoretical value or population mean, s is the standard deviation, and n is the group size or sample size.

Independent Two Sample t-test t-statistic for two groups A and B is computed as follows:

$$t = \frac{\text{mean}(A) - \text{mean}(B)}{\sqrt{\frac{(N_1 - 1)s_1^2 + (N_2 - 1)s_2^2}{N_1 + N_2 - 2} \left(\frac{1}{N_1} + \frac{1}{N_2} \right)}} \quad (2.50)$$

Here, mean(A) and mean(B) are for two different samples. N1 and N2 are sample sizes of two groups A and B. s² is the variance of the two samples and the degree of freedom is given as N1 + N2 - 2. Then, t-statistic is compared with the t-critical value.

Paired t-test It is used to evaluate the hypothesis before and after intervention. The fact is that these samples are not independent. For example, consider the case of an effect of medication for a diabetic patient. The sequence is that first the sugar is tested, then the medication is done, and again the sugar test is conducted to study the effect of medication. In short, in paired t-test, the data is taken from the same subject twice. In an unpaired t-test, the samples are taken independently. In this only one group is involved. The t-statistic is computed as:

$$t = \frac{m - \mu}{\frac{s}{\sqrt{n}}}$$

Here, t is t-statistic, m is the mean of the group, m is the theoretical value or population mean, s is the standard deviation, and n is the group size or sample size.

Chi-Square Test

Chi-Square test is a non-parametric test. The goodness-of-fit test statistics follows a Chi-Square distribution under null hypothesis and measures the statistical significance between observed frequency and expressed frequency, and each observation is independent of each other and follows normal distribution. This

$$\chi^2 = \sum_{i=1}^n \frac{(O_i - E_i)^2}{E_i} \quad (2.51)$$

comparison is used to calculate the value of the Chi-Square statistic as:

Here, E is the expected frequency, O is the observed frequency and the degree of freedom is C - 1, where, C is number of categories. The Chi-Square test allows us to detect the duplication of data and helps to remove the redundancy of values.

FEATURE ENGINEERING AND DIMENSIONALITY REDUCTION TECHNIQUES

Features are attributes. Feature engineering is about determining the subset of features that form an important part of the input that improves the performance of the model, be it classification or any other model in machine learning.

Feature engineering deals with two problems – Feature Transformation and Feature Selection. Feature transformation is extraction of features and creating new features that may be helpful in increasing performance. For example, the height and weight may give a new attribute called Body Mass Index (BMI).

Feature subset selection is another important aspect of feature engineering that focuses on selection of features to reduce the time but not at the cost of reliability.

The features can be removed based on two aspects:

1. Feature relevancy – Some features contribute more for classification than other features. For example, a mole on the face can help in face detection than common features like nose. In simple words, the features should be relevant.
2. Feature redundancy – Some features are redundant. For example, when a database table has a field called Date of birth, then age field is not relevant as age can be computed easily from date of birth.

So, the procedure is:

1. Generate all possible subsets
2. Evaluate the subsets and model performance
3. Evaluate the results for optimal feature selection

Filter-based selection uses statistical measures for assessing features. In this approach, no learning algorithm is used. Correlation and information gain measures like mutual information and entropy are all examples of this approach.

Wrapper-based methods use classifiers to identify the best features. These are selected and evaluated by the learning algorithms. This procedure is computationally intensive but has superior performance.

Stepwise Forward Selection

This procedure starts with an empty set of attributes. Every time, an attribute is tested for statistical significance for best quality and is added to the reduced set. This process is continued till a good reduced set of attributes is obtained.

Stepwise Backward Elimination

This procedure starts with a complete set of attributes. At every stage, the procedure removes the worst attribute from the set, leading to the reduced set.

Principal Component Analysis

The idea of the principal component analysis (PCA) or KL transform is to transform a given set of measurements to a new set of features so that the features exhibit high information packing properties.

This leads to a reduced and compact set of features. Consider a group of random vectors of the form:

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

$$m_i = E[x_i]$$

The mean vector of the set of random vectors is defined as:

The operator E refers to the expected value of the population. This is calculated theoretically using the probability density functions (PDF) of the elements x_i and the joint probability density functions between the elements x_i and x_j . From this, the covariance matrix can be calculated as:

$$m_x = \frac{1}{M} \sum_{k=1}^M x_k \quad (2.53)$$

$$A = \frac{1}{M} \sum_{k=1}^M x_k x_k^T - m_x m_x^T \quad (2.54)$$

This covariance matrix is real and symmetric. If e_i and λ_i (where, $i = 1, 2, \dots, n$) be the set of eigen vectors and corresponding eigen values of the covariance matrix, the eigen values can be arranged in a descending order so that $\lambda_i \geq \lambda_{i+1}$ for $i = 1, 2, \dots, n - 1$. The corresponding eigen vectors are calculated. Based on this, the transform kernel is constructed. Let the transform kernel be A. Then, the matrix rows are formed from the eigen vectors of the covariance matrix.

The mapping of the vectors x to y using the transformation can now be described as:

$$y = A(x - m_x) \quad (2.55)$$

This transform is also called as Karhunen-Loeve or Hotelling transform. The original vector

$$x = A^T y + m_x \quad (2.56)$$

x can now be reconstructed as follows:

$$x = A_k^T y + m_x \quad (2.57)$$

If K largest eigen values are used, the recovered information would be:

The PCA algorithm is as follows:

1. The target dataset x is obtained
2. The mean is subtracted from the dataset. Let the mean be m . Thus, the adjusted dataset is $X - m$. The objective of this process is to transform the dataset with zero mean.
3. The covariance of dataset x is obtained. Let it be C .
4. Eigen values and eigen vectors of the covariance matrix are calculated.
5. The eigen vector of the highest eigen value is the principal component of the dataset. The eigen values are arranged in a descending order. The feature vector is formed with these eigen vectors in its columns.
Feature vector = {eigen vector1, eigen vector2, ..., eigen vectorn}
6. Obtain the transpose of feature vector. Let it be A .
7. PCA transform is $y = A \times (x - m)$, where x is the input dataset, m is the mean, and A is the transpose of the feature vector.

$$\text{Original data } (f) = \{(A)^{-1} \times y\} + m \quad (2.58)$$

$$= \{(A)^T \times y\} + m \quad (2.59)$$

The original data can be retrieved using the formula given below:

The new data is a dimensionally reduced matrix that represents the original data.

Figure 2.15.

The scree plot indicates that only 6 out of 246 attributes are important.

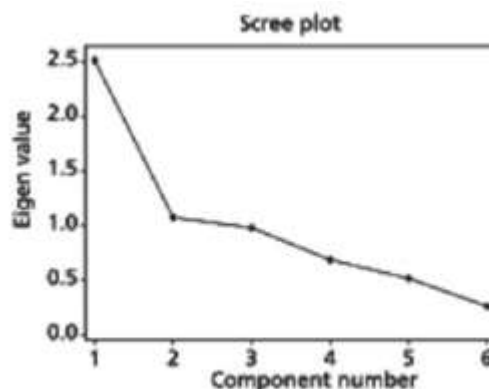


Figure 2.15: Scree Plot

one can infer the relevance of the attributes. The scree plot indicates that the first attribute is more important than all other attributes.

Linear Discriminant Analysis

Linear Discriminant Analysis (LDA) is also a feature reduction technique like PCA. The focus of LDA is to project higher dimension data to a line (lower dimension data). LDA is also used to classify the data. Let there be two classes, $c1$ and $c2$. Let $m1$ and $m2$ be the mean of the patterns of two classes. The mean of the class $c1$ and $c2$ can be computed as:

$$\mu_1 = \frac{1}{N_1} \sum_{x_i \in c_1} x_i \text{ and } \mu_2 = \frac{1}{N_2} \sum_{x_i \in c_2} x_i$$

The aim of LDA is to optimize the function:

$$J(V) = \frac{V^T \sigma_B V}{V^T \sigma_W V} \quad (2.60)$$

where, V is the linear projection and σ_B and σ_W are class scatter matrix and within scatter matrix, respectively. For the two-class problem, these matrices are given as:

$$\sigma_B = N_1(\mu_1 - \mu)(\mu_1 - \mu)^T + N_2(\mu_2 - \mu)(\mu_2 - \mu)^T \quad (2.61)$$

$$\sigma_W = \sum_{x_i \in c_1} (x_i - \mu_1)(x_i - \mu_1)^T + \sum_{x_i \in c_2} (x_i - \mu_2)(x_i - \mu_2)^T \quad (2.62)$$

The maximization of $J(V)$ should satisfy the equation:

$$\sigma_B V = \lambda \sigma_W V \text{ or } \sigma_W^{-1} \sigma_B V = \lambda V \quad (2.63)$$

As $\sigma_B V$ is always in the direction of $(\mu_1 - \mu_2)$, V can be given as:

$$V = \sigma_W^{-1}(\mu_1 - \mu_2) \quad (2.64)$$

Let $V = [v_1, v_2, \dots, v_d]$ be the generalized eigen vectors of σ_B and σ_W , where, d is the largest eigen values as in PCA. The transformation of x is then given as:

$$y = V_d^T x \quad (2.65)$$

Like in PCA, the largest eigen values can be retained to have projections.

Singular Value Decomposition

Singular Value Decomposition (SVD) is another useful decomposition technique. Let A be the matrix, then the matrix A can be decomposed as:

$$A = U S V^T \quad (2.66)$$

Here, A is the given matrix of dimension $m \times n$, U is the orthogonal matrix whose dimension is $m \times m$, S is the diagonal matrix of dimension $n \times n$, and V is the orthogonal matrix. The procedure for finding decomposition matrix is given as follows:

1. For a given matrix, find AA^T
2. Find eigen values of AA^T
3. Sort the eigen values in a descending order. Pack the eigen vectors as a matrix U .
4. Arrange the square root of the eigen values in diagonal. This matrix is diagonal matrix, S .
5. Find eigen values and eigen vectors for $A^T A$. Find the eigen value and pack the eigen vector as a matrix called V .